

VIKAS KUMAR

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EXPERIENCE

Lightbeam.ai | Machine Learning Lead - Founding Engineer

Bangalore, Dec'20 - Present

- **Architected multi-stage information extraction pipeline** using deep learning models and distributed processing (Celery, Kafka) to handle 2-3TB daily, achieving 93% F1 score across unstructured text, PDFs, and scanned images.
- **Engineered production-scale semantic search system** enabling real-time document similarity searches across 20M+ documents with sub-600ms, leveraging Qdrant vector DB for dense embedding storage and retrieval.
- **Built document classification system** processing unstructured data (PDFs, images, Office files) across 16 categories, achieving 89% accuracy with BERT models optimized via ONNX for 20% faster inference.
- **Leveraged and deployed** quantized Llama 3 via Ollama for data labeling, keyword-based template creation, and topic classification across large-scale document clusters.
- **Progressed from Founding ML Engineer to ML Lead**, scaling product from concept to production while mentoring 4+ engineers in ML experimentation and deployment, collaborating cross-functionally with product and customer teams.

Deloitte Consulting USI | Consultant – Applied AI (Strategy & Analytics)

Gurgaon, May'19 - Nov'20

- **Led Fortune 500 client engagements** across insurance, healthcare, and media industries, translating complex business requirements into ML solutions generating \$2M+ in measurable ROI
- **Designed and deployed intelligent NLP solutions** including email response classification system using XGBoost & BERT achieving 81% F1 across 20+ categories automating 60% of responses.
- **Spearheaded ML pipeline development** for strategic market intelligence, encompassing data extraction, feature engineering, and building predictive models (XGBoost, 67% F1) to identify biologic market size and high-potential HCP.

EXL Analytics | Data Scientist – Product Development Lab

Gurgaon, Aug'17 - Apr'19

- **Built email classification system** using ULMFiT transfer learning to automatically classify email intent based on subject and body content, delivering 89% accuracy and outperforming XGBoost by 8%.
- **Implemented computer vision algorithms** using Mask R-CNN for seller/buyer information extraction from bills of lading and developed custom table detection models (DeepDeSRT) for document structure recognition.
- **Built in-house speech recognition platform** implementing and comparing DeepSpeech and WaveNet architectures, achieving 21% and 34% WER respectively for audio-to-text transcription.

EDUCATION

Indian Institute of Technology, Kharagpur

2012 - 2017

Dual Degree (B.Tech & M.Tech) in Agricultural & Food Engineering; GPA: 7.76

TECHNICAL SKILLS

Languages & Tools

Python, PyTorch, HuggingFace, MongoDB, Redis and Qdrant

ML Domains

Deep Learning, NLP, Computer Vision, LLM and Machine Learning

Deployment and MLOps

Docker, Kubernetes, Kafka, Celery, FastAPI and DVC

INTERNSHIPS

- **Research Assistant, Kansas State University, USA** (May-Jul 2015): Developed predictive model for spray coating optimization, reducing coefficient of variation by 21% through experimental design and statistical analysis.
- **Decision Science Intern, Quantiphi Analytics** (May-Jul 2016): Built predictive models for TV show performance analysis using feature engineering on anchorship effects, historical performance, and TRPs, achieving 10% MAPE.

ADDITIONAL

- **Technical Leadership:** Created NLP training curriculum for training 200+ engineers; Kaggle Days speaker & mentor
- **Kaggle Competitions:** Multiple silver medals; Top 5% in Quora NLP (173/4000+), Computer Vision Data Science Bowl (172/3634), Ad Fraud Detection (193/3900+)
- **Professional Recognition:** Multiple Applause Awards at Deloitte USI for leading high-impact client engagements and driving technical excellence